

Multilingual Classification of Urgent Disaster Response Messages from Social Media

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Abstract—During natural disasters, many people use social media to ask for help, share updates, and coordinate relief work. However, the number of posts is usually very large, and only a small part of them may need urgent action. This makes it difficult for response teams to manually find the most important messages. In this work, we build a Natural Language Processing (NLP) pipeline to classify disaster-related social media messages into five classes: Situational Awareness, Critical Rescue, Volunteering and Donations, Irrelevant, and Resource Requests. We use the HumAID dataset and handle class imbalance using two methods: synthetic data generation with a Large Language Model (LLM) for the minority class, and class weighting during training. We compare traditional Machine Learning models such as Naive Bayes, Logistic Regression, and Linear SVM with four Transformer models: mBERT, XLM-RoBERTa, IndicBERT, and MuRIL. We also test Llama 3.3 70B as a zero-shot and few-shot classifier. To study cross-lingual transfer, we created a small Malayalam Twitter test set and a Manglish version of the same data. All models are trained only on English and tested on Malayalam and *Manglish* without Malayalam fine-tuning. Fine-tuned XLM-RoBERTa gives the best English result with a macro-F1 score of 0.778. MuRIL performs best on Malayalam with macro-F1 of 0.409 and on *Manglish* with macro-F1 of 0.188. These results show that multilingual Transformers can transfer useful meaning across languages and scripts, even when no target-language training data is used.

I. INTRODUCTION

Social media platforms such as X (formerly Twitter), Telegram, and WhatsApp are often used as communication channels during natural disasters. People post requests for food, medical help, rescue, shelter, and other urgent needs. At the same time, many users also post news updates, safety messages, donation details, or general support messages. Because of this, emergency teams have to search through a large number of posts to find the few that need immediate action. Manual checking is slow and difficult in such situations. For example, agencies such as the National Disaster Response Force (NDRF) cannot read every post during a flood or cyclone. A system that can automatically classify social media messages can help responders focus on the most useful and urgent posts first. Simple keyword matching is not enough for this task. A word like *water* may appear in different types of messages. It may be part of a rescue request, a resource request, a flood update, or even a thank-you message. Therefore, the model must understand the meaning of the full sentence, not just the presence of specific words. This problem becomes harder in India because people use many languages and scripts online. During Kerala floods, for example, posts may appear in Malayalam script, English, or *Manglish*, where Malayalam is written using English letters, basically transliteration. Traditional English-based NLP methods do not handle this well.

In this work, we treat the task as a five-class text classification problem based on the HumAID taxonomy [7]. The five classes are Situational Awareness, Critical Rescue, Volunteering and

Donations, Irrelevant, and Resource Requests. Our main goal is to check how well different models can classify urgent disaster messages and how well English-trained models transfer to Malayalam and Manglish. Our main contributions are:

- We built an end-to-end pipeline that includes label mapping, text cleaning, synthetic data generation, and model training.
- We compared traditional Machine Learning models with fine-tuned multilingual Transformer models on the same task.
- We tested zero-shot cross-lingual performance on a manually annotated Malayalam Twitter dataset and a Manglish version of the same dataset.

II. DATA AND PREPROCESSING

A. Dataset Description

The main dataset used in this work is the HumAID dataset [7]. It is a human-annotated collection of disaster-related tweets from 17 natural disasters. The dataset contains about 74,000 tweets. The original split has 51,952 training tweets, 7,564 development tweets, and 14,711 test tweets. Table I gives one cleaned example from each target class.

We also created a small regional test set for Malayalam. It contains 478 manually annotated Malayalam disaster tweets collected from Twitter during the 2018 and 2019 Kerala flood events. We also created a Manglish version of the same tweets. Personal details such as phone numbers and names were masked wherever needed before annotation. The Malayalam tweets were annotated using Label Studio. Each tweet was assigned one of the five labels: Critical Rescue, Resource Requests, Volunteering and Donations, Situational Awareness, and Irrelevant. The annotation interface is shown in Figure 1.



Fig. 1. Label Studio interface used for manual annotation of Malayalam disaster tweets.

B. Label Harmonization

The HumAID dataset originally has 11 detailed labels. For this project, we mapped those labels into our five target classes. During this step, we found an important issue in an earlier version of our pipeline. The `sympathy_and_support` label

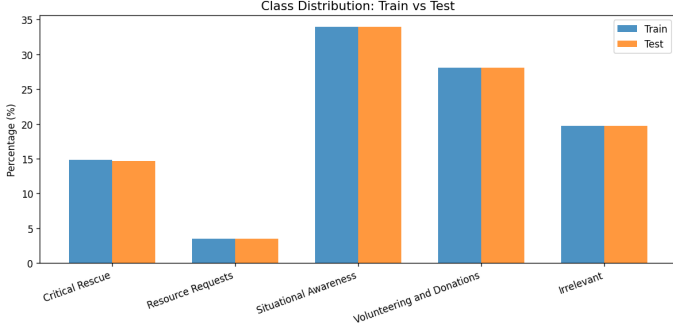


Fig. 5. Aggregate class distribution on the HumAID dataset after label harmonization.

prompt asked the model to create short, realistic social-media-style disaster posts. It included instructions to use hashtags, urgent words, abbreviations such as *plz*, *u*, and *4*, and small typos. To reduce repetition, we used five different real training examples as seed examples and rotated them across calls. We also used a near-duplicate filter to remove generated tweets that were too similar to earlier ones. Only training examples were used as seeds, so no development or test data leaked into the generation process. After filtering, we added 2,500 synthetic *Resource Requests* tweets.

The training set changed as follows:

- Original training set: 51,952 rows
- Augmented training set: 54,452 rows
- Resource Requests before: 1,815 training rows (3.5%)
- Resource Requests after: 4,315 training rows (7.9%)

The following real example shows the type of message the model should learn to identify:

“PLEASE HELP us with food & water, 200 ppl stranded in shelter, opp to govt school, khadki, pune. dont leave us hungry pls 90234 56789 #MaharashtraFloods @NGOIndia”

2) Algorithmic class weighting. For the supervised models, we also used class weighting. In scikit-learn models, we used `class_weight='balanced'` where possible. For transformer models, we used equivalent sample weighting during fine-tuning. This gives more importance to minority classes during training.

Both methods were used together in the main experiments. Section V separately checks how much the synthetic data helped.

III. TRADITIONAL MACHINE LEARNING BASELINES

A. Model Setup

We first tested traditional Machine Learning models to set a baseline. We trained Multinomial Naive Bayes, Logistic Regression, and Linear SVM. These models used TF-IDF features with unigrams and bigrams. The feature size was capped at 30,000, with `min_df=3`, `max_df=0.95`, and sub-linear term-frequency scaling. MaxAbs scaling was used so that the sparse feature matrix remained sparse. Class balancing was applied using sample weights for Naive Bayes and `class_weight='balanced'` for Logistic Regression and Linear SVM.

B. English Test Set Performance

Table II shows the macro-averaged performance of the three baseline models on the English HumAID test set.

TABLE II
BASELINE PERFORMANCE ON THE ENGLISH HUMAID TEST SET

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Logistic Regression	0.790	0.745	0.754	0.749
Linear SVM	0.759	0.714	0.709	0.711
Naive Bayes	0.746	0.700	0.682	0.688

Logistic Regression performs best among the three baselines, with a macro-F1 score of 0.749. This gives a useful starting point for comparison. Figure 6 compares recall across the three baseline models. Recall is important in this task because missing a real call for help can be more serious than wrongly flagging a less urgent message.

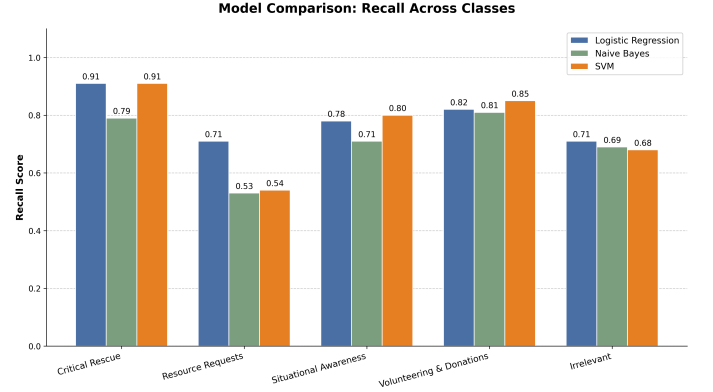


Fig. 6. Recall comparison of the three traditional machine learning baselines.

C. The Multilingual Roadblock

TF-IDF represents text using the exact words that appear in the training data. This means that an English-trained TF-IDF model has no useful representation for Malayalam words. It cannot understand that a Malayalam sentence may have the same meaning as an English sentence. Because of this, traditional baselines are expected to perform poorly on Malayalam. This is why we also test multilingual Transformer models, which use shared sub-word representations across languages.

IV. TRANSFORMER MODELS

A. Reality Check: Untrained Transformers

Before fine-tuning, we tested the base Transformer models directly on the English test set. This was done to check whether pre-training alone was enough for this classification task. Table III shows that the untrained models perform much worse than the traditional baselines, with macro-F1 scores below 0.10. This result is expected. Pre-training teaches these models general language patterns, but it does not teach them our five disaster-response labels. The classification head is randomly initialized and must be trained for this task.

B. Fine-Tuning Setup

We fine-tuned four multilingual Transformer models:

- **mBERT** [4]: a standard multilingual BERT model used as a common baseline.
- **XLM-RoBERTa** [2]: a strong multilingual model trained on large-scale multilingual data.
- **MuRIL** [5]: a model trained for Indian languages and transliterated Indian text, making it useful for Malayalam and Manglish.

TABLE III
UNTRAINED TRANSFORMERS VS. TRADITIONAL BASELINES (ENGLISH TEST SET)

Model	Accuracy	Macro F1
Logistic Regression (baseline)	0.790	0.749
Linear SVM (baseline)	0.759	0.711
Naive Bayes (baseline)	0.746	0.688
Base XLM-RoBERTa (untrained)	0.281	0.088
Base IndicBERT (untrained)	0.147	0.052
Base mBERT (untrained)	0.048	0.034
Base MuRIL (untrained)	0.035	0.015

- **IndicBERT** [3]: a lightweight model focused on Indian languages.

All four models were fine-tuned on the augmented English training set using the same hyperparameters. These settings are shown in Table IV. We used the HuggingFace Trainer API with cross-entropy loss and balanced sample weights.

TABLE IV
FINE-TUNING HYPERPARAMETERS (SHARED ACROSS ALL FOUR TRANSFORMERS)

Hyperparameter	Value
Learning rate	2e-5
Batch size (train / eval)	32 / 64
Epochs	3
Max sequence length	128 tokens
Optimizer	AdamW
Weight decay	0.01
Warmup ratio	0.1
LR schedule	Linear decay
Random seed	42

Table V shows the training and evaluation efficiency of the models. XLM-RoBERTa had the fastest evaluation time and the lowest evaluation loss. mBERT had the lowest training loss but was slower during evaluation.

TABLE V
TRAINING AND EVALUATION EFFICIENCY

Metric	MuRIL	IndicBERT	mBERT	XLM-R
Eval Loss	0.503	0.513	0.549	0.486
Eval Runtime (s)	5.04	4.78	5.07	4.15
Eval Samples/sec	1498.8	1582.0	1489.0	1821.1
Train Loss	0.524	0.466	0.407	0.438
Train Runtime (s)	414.4	280.8	381.3	343.7
Train Samples/sec	394.4	582.0	428.6	475.4

C. Final English Performance

After fine-tuning, the Transformer models clearly perform better than the traditional baselines. Table VI shows all seven supervised models sorted by macro-F1. Fine-tuned XLM-RoBERTa performs best with a macro-F1 score of 0.778. MuRIL, mBERT, and IndicBERT are also close. The best traditional baseline, Logistic Regression, remains lower than all fine-tuned Transformer models.

Figures 7 and 8 compare accuracy and recall across all models. Table VI is kept compact because macro recall was not stored for every model checkpoint in this table. Therefore, macro precision cannot be reliably calculated from only accuracy and macro-F1. The recall comparison is especially important because recall shows how well the model finds actual urgent messages.

TABLE VI
FINAL PERFORMANCE: FINE-TUNED TRANSFORMERS VS. TRADITIONAL BASELINES (ENGLISH TEST SET)

Model	Type	Accuracy	Macro F1
Fine-Tuned XLM-RoBERTa	Transformer	0.848	0.778
Fine-Tuned MuRIL	Transformer	0.831	0.773
Fine-Tuned mBERT	Transformer	0.829	0.769
Fine-Tuned IndicBERT	Transformer	0.814	0.753
Logistic Regression	ML Baseline	0.790	0.749
Linear SVM	ML Baseline	0.759	0.711
Naive Bayes	ML Baseline	0.746	0.688

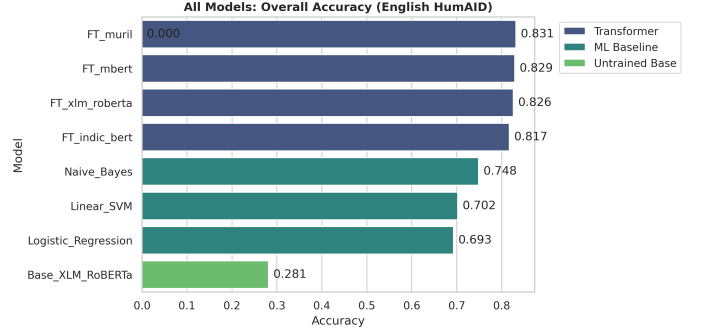


Fig. 7. Accuracy comparison across all models on the English HumAID test set.

D. Per-Class Error Analysis

Figure 9 shows per-class recall for all supervised models. Even the best Transformer models perform worse on *Resource Requests* compared to larger classes. This is expected because *Resource Requests* is the smallest class. It is also close in meaning to some *Situational Awareness* tweets, especially when both mention relief work, supplies, or affected areas.

Another reason for errors is that some tweets naturally fit more than one class. For example, a tweet asking for boat rescue may also mention that a road is flooded. In our setup, each tweet can have only one label, so the model is forced to choose one class. This can reduce recall and make some errors harder to interpret.

V. SYNTHETIC DATA ABLATION

To check whether the synthetic data helped, we trained the three traditional baselines in two settings. In the first setting, we used only the original training data. In the second setting, we used the augmented data with 2,500 extra synthetic *Resource Requests* tweets. All other settings were kept the same. The results are shown in Table VII.

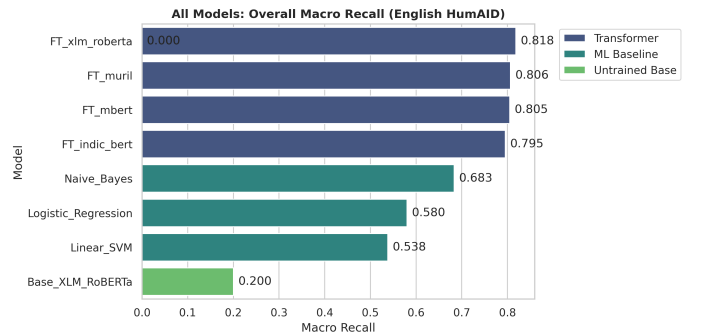


Fig. 8. Recall comparison across all models on the English HumAID test set.

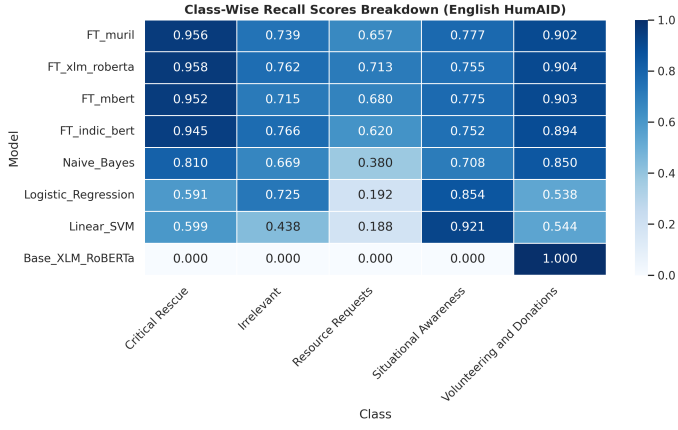


Fig. 9. Per-class recall heatmap across all models on the English test set.

TABLE VII
ABLATION: REAL-ONLY VS. AUGMENTED TRAINING (ENGLISH TEST SET)

Model	Macro F1		F1 on Resource Requests		Test Accuracy	
	Real	Aug.	Real	Aug.	Real	Aug.
Naive Bayes	0.652	0.662	0.51	0.56	0.744	0.748
Logistic Regression	0.576	0.589	0.43	0.49	0.686	0.693
Linear SVM	0.568	0.579	0.42	0.48	0.695	0.702

The synthetic data gives a small but consistent improvement. Macro-F1 improves by about 0.01 to 0.013, and the F1 score for *Resource Requests* improves by about 0.05 to 0.06. This means the generated tweets helped the model learn the minority class better. However, the gain is not very large. This is reasonable because synthetic tweets cannot fully replace real disaster tweets, which are usually more varied and noisy.

VI. PROMPTED LLM CLASSIFIER

We also tested Llama 3.3 70B through the Groq API as a prompted classifier. For this experiment, we sampled 50 English test tweets, with 10 tweets from each class. The model was asked to return exactly one label for each tweet. We tested three prompt types: zero-shot with detailed label descriptions, zero-shot with only label names, and one-shot with one example per class. We also ran the detailed zero-shot setup twice with different random seeds to check variation. The results are shown in Table VIII.

TABLE VIII
LLAMA 3.3 70B CLASSIFICATION ACCURACY ON ENGLISH TEST SUBSETS

Prompting Strategy	Seed	Accuracy
Zero-shot, detailed labels	42	0.54
Zero-shot, minimal labels	99	0.50
One-shot, detailed labels	123	0.62
Zero-shot, detailed labels	200	0.56

The best LLM result is 0.62 accuracy with one-shot prompting. This is lower than the fine-tuned Transformer models. The result suggests that LLM prompting can work to some extent, but task-specific fine-tuning is still better for this classification problem. One-shot prompting improves over zero-shot because the model gets one example for each class. Detailed label descriptions also help slightly compared to only giving label names. Since this experiment uses only 50 tweets, the results should be treated as indicative rather than final. We tested the LLM only on English because Llama 3.1/3.3 has limited

Malayalam ability, and that would make the cross-lingual comparison unfair.

VII. CROSS-LINGUAL EVALUATION

A. Motivation: Zero-Shot Transfer

The fine-tuned Transformer models were trained only on English tweets. They were not trained on Malayalam sentences. We still test them on Malayalam because this is close to a real disaster situation. When a disaster happens in a new region, there may not be enough time to collect and label local-language training data. A useful system should be able to transfer some knowledge from English to other languages.

B. Semantic Alignment Visualization

Multilingual Transformer models can work across languages because they place text from different languages into a shared embedding space. In simple terms, sentences with similar meanings should have similar representations, even if they are written in different languages. To check this, we passed one English sentence and one Malayalam sentence with the same meaning through the fine-tuned MuRIL model. Their embeddings had a cosine similarity of 0.865, where 1.0 means perfect similarity. Figure 10 shows the sentence pair. Figure 11 shows a t-SNE plot of MuRIL embeddings across our test sets. Similar-intent tweets appear closer together even when the language changes.

English: 'We need urgent rescue from the flood, please help!'

Malayalam: 'പ്രളയത്തിൽ നിന്നും തങ്ങളെ അടിയന്തരമായി രക്ഷിക്കണം, ദയവായി സഹായിക്കുക!'

Fig. 10. An English-Malayalam sentence pair that MuRIL maps to closely aligned embeddings (cosine similarity 0.865).

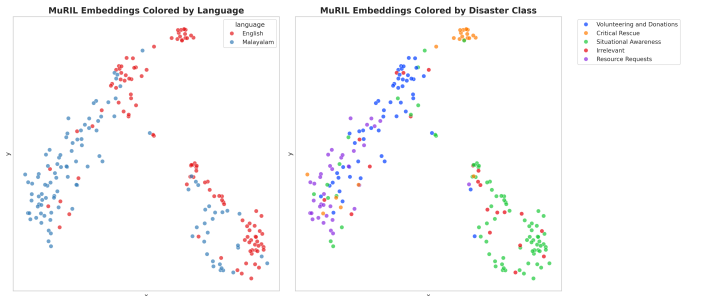


Fig. 11. t-SNE projection of MuRIL embeddings on the cross-lingual test sets. Same-intent clusters form across languages.

C. Results on Native Malayalam

Table IX shows zero-shot performance on the 108-tweet Malayalam test set. MuRIL performs best with a macro-F1 score of 0.409. IndicBERT is close behind with 0.404. These two models perform better than XLM-RoBERTa and mBERT, which is expected because MuRIL and IndicBERT are more focused on Indian languages. Traditional baselines perform poorly because they were trained on English TF-IDF features.

Figures 12 and 13 show accuracy and recall on the Malayalam test set.

TABLE IX
ZERO-SHOT PERFORMANCE ON THE NATIVE MALAYALAM TEST SET

Model	Type	Accuracy	Precision	Recall	Macro F1
FT MuRIL	Transformer	0.542	0.402	0.416	0.409
FT IndicBERT	Transformer	0.514	0.394	0.415	0.404
FT XLM-RoBERTa	Transformer	0.449	0.290	0.342	0.314
FT mBERT	Transformer	0.327	0.224	0.254	0.238
Naive Bayes	ML Baseline	0.224	0.105	0.209	0.140
Base XLM-RoBERTa	Untrained	0.280	0.056	0.200	0.088
Logistic Regression	ML Baseline	0.150	0.054	0.203	0.085
Linear SVM	ML Baseline	0.224	0.048	0.192	0.077

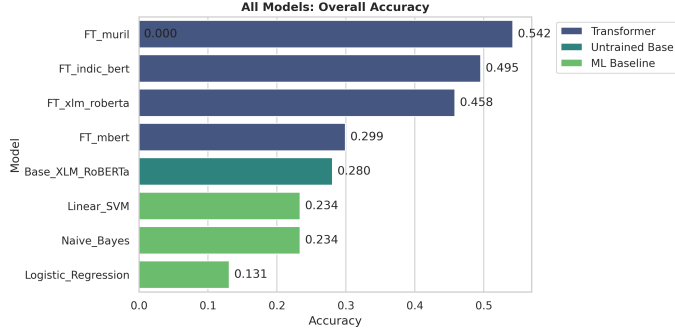


Fig. 12. Zero-shot accuracy on the native Malayalam test set.

D. Results on Transliterated Manglish

Manglish is Malayalam written using English letters. It is common in social media posts, especially on mobile phones. It is also difficult for models because there is no fixed spelling. The same Malayalam word can be written in many different ways. Manglish posts may also mix English and Malayalam words in the same sentence. Table X shows the results on the Manglish test set.

TABLE X
ZERO-SHOT PERFORMANCE ON THE TRANSLITERATED MANGLISH TEST SET

Model	Type	Accuracy	Precision	Recall	Macro F1
FT MuRIL	Transformer	0.346	0.151	0.249	0.188
FT mBERT	Transformer	0.262	0.128	0.240	0.167
Naive Bayes	ML Baseline	0.178	0.128	0.173	0.147
Logistic Regression	ML Baseline	0.178	0.067	0.227	0.103
Linear SVM	ML Baseline	0.215	0.058	0.190	0.089
FT XLM-RoBERTa	Transformer	0.140	0.058	0.188	0.089
Base XLM-RoBERTa	Untrained	0.280	0.056	0.200	0.088
FT IndicBERT	Transformer	0.159	0.054	0.223	0.087

All models perform worse on Manglish than on native Malayalam. MuRIL still gives the best result, which supports the idea that its transliterated Indian-language pre-training helps. The

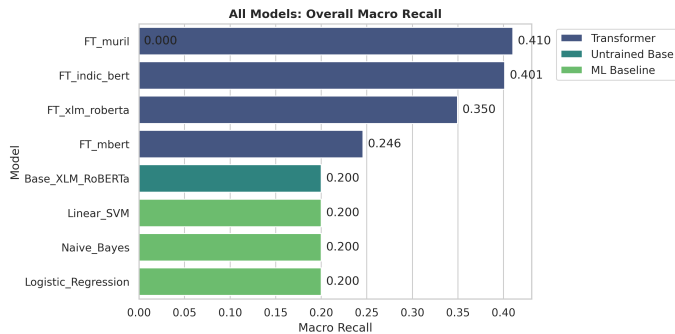


Fig. 13. Zero-shot recall on the native Malayalam test set.

traditional models perform slightly better on Manglish than on Malayalam because some Manglish words use English letters and may overlap with the English TF-IDF vocabulary. However, their overall performance is still low.

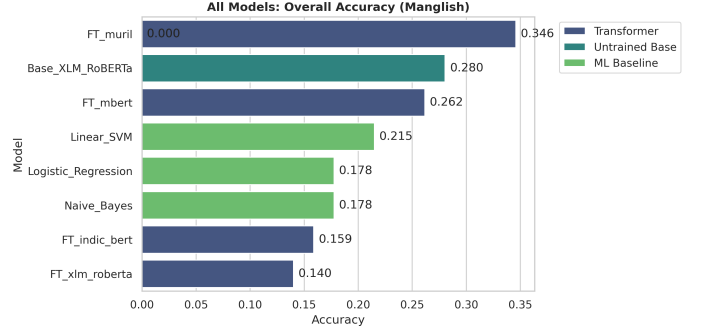


Fig. 14. Zero-shot accuracy on the transliterated Manglish test set.

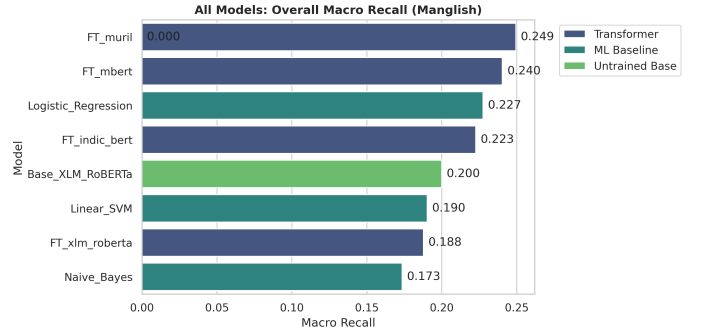


Fig. 15. Zero-shot recall on the transliterated Manglish test set.

VIII. LIMITATIONS

This work has some limitations. **Synthetic data cannot replace real data.** The LLM-generated *Resource Requests* tweets improve performance, but the improvement is small. Generated tweets are often cleaner and less varied than real disaster tweets. More real annotated examples would likely help this class more. **The task is single-label.** In real life, one tweet may belong to more than one class. For example, a message may ask for rescue and also report a flooded road. Our setup forces each tweet to have only one label, which can make the problem harder and can hide useful information. **The Malayalam and Manglish test sets are small.** Each set has 108 manually annotated tweets. This is enough to show general trends, but not enough for very precise performance estimates. **The LLM classifier test is also small.** The Llama 3.3 experiments use only 50 tweets because of API cost. Therefore, the results should be seen as a small comparison rather than a full evaluation. **Class imbalance still remains.** Even after synthetic data generation and class weighting, *Resource Requests* remains the hardest class. This shows that imbalance is still an important issue in disaster-response classification.

IX. FUTURE WORK

There are several possible ways to extend this work:

- **Multi-label classification.** Future models can allow one tweet to have more than one label. This would better match real disaster messages.

- **More Indian languages.** The same pipeline can be tested on Hindi, Tamil, Telugu, Bengali, and other Indian languages, including their transliterated forms.
- **Use of images and videos.** Disaster tweets often contain images or videos. A multimodal model could use both text and visual information.
- **Confidence scores.** Models with reliable confidence scores can help responders decide which predictions need manual checking.

X. CONCLUSION

In this project, we built a multilingual classification pipeline for disaster-response tweets. We tested it on English, Malayalam, and Manglish data. The results show three main points. First, fine-tuned multilingual Transformer models perform much better than traditional Machine Learning baselines and prompted LLM classification on the English dataset. XLM-RoBERTa gives the best English macro-F1 score of 0.778. Second, zero-shot transfer to Malayalam is possible. MuRIL reaches a macro-F1 score of 0.409 on Malayalam without any Malayalam fine-tuning. Third, Manglish is harder than native Malayalam, but MuRIL still performs best with a macro-F1 score of 0.188. The synthetic data experiment shows that LLM-generated examples can help the minority *Resource Requests* class, but only to a limited extent. Overall, the results suggest that multilingual Transformer models can be useful for disaster-response message classification in Indian language settings. However, more annotated regional data and better handling of multi-label tweets would be needed before such a system is used in real emergency response.

REPRODUCIBILITY

All code, trained baselines, the processed HumAID dataset, the manually annotated Malayalam and Manglish test sets, and the synthetic *Resource Requests* data are available in the project repository. Random seeds are fixed throughout the pipeline. Raw HumAID data is not redistributed due to licensing; a download script is provided.

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